

Markov Random Fields with Geographic Seeds

Bayesian Estimation on Hierarchical Hidden Field Model

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1. Introduction

The Potts Model

- Spatial models on regular lattices (pixels) often employ **Markov Random Fields (MRF)**.
- The **Potts Model** is the classical standard for capturing local neighborhood dependence.

The Geographic Limitation

- Many real images and textures possess geographic patterns we call “**seeds**” that may attract or repel colors.
- **Our Proposal:** Incorporate geographic seeds that exert radial influence on the classes.
- **Objective:** Propose a reparameterized model that describe the inclusion of centroids/seeds, and validate its inference under a hidden model.

2. The Proposal

Potts as is

- **Energy Function:** For the traditional Potts Model, we define a spatial field Z where the local probability of pixel i taking class k is proportional to $\exp(\eta_i(k))$:

$$\eta_i(k) = \alpha \sum_{j \in N(i)} \mathbb{1}(Z_j = k) + \beta \mathbb{1}(k = 0)$$

- **Parameters Meaning:**
 - $N(i)$: Pixel i neighbors.
 - α (**Uniformity**): Local attraction between neighboring pixels (smoothing).
 - β (**Global Abundance**): Force favoring/penalizing the background (class 0).

Potts Reparameterized with Seeds

- **Enhanced Local Energy Function:** To address the isotropic limitation of the Potts model, we introduce a few elements to handle the geographic centrality bias in our local energy function:

$$\eta_i(k) = \alpha \sum_{j \in N(i)} 1(Z_j = k) + \beta 1(k = 0) + \sum_s \frac{\delta_s}{1 + d(i, s)} 1(c_s = k)$$

- **Parameters Meaning:**
 - α (**Uniformity**): Local attraction between neighboring pixels (smoothing).
 - β (**Global Abundance**): Force favoring/penalizing the background (class 0).
 - δ_s (**Centrality**): Radial attraction/repulsion of the seed s .
 - $d(i, s)$: Distance between pixel i and seed s .
 - c_s : Class/color related to the seed.

Potts Reparameterized with Seeds

- **Enhanced Local Energy Function:**

$$\eta_i(k) = \alpha \sum_{j \in N(i)} 1(Z_j = k) + \beta 1(k = 0) + \sum_s \frac{\delta_s}{1 + d(i, s)} 1(c_s = k)$$

- **Enhanced Global Energy Function:**

$$U(\omega) = \alpha \sum_{\{i,j\} \in E} \sum_{k \in S} s_{i,k} s_{j,k} + \beta 1(k = 0) + \sum_s \frac{\delta_s}{1 + d(i, s)} 1(c_s = k)$$

The Threshold of the “Area of Influence”

- **Equilibrium of Forces:** The local seed attraction competes against the global abundance of the background class.
- **Null Boundary:** Under our model, ignoring neighborhood smoothing, they cancel out when:

$$\eta_i(0) = \eta_i(k) \implies d_{i,s} = \frac{\delta_s}{\beta} - 1$$

- **Geographic Transition:**
 - **Within radius d_i :** The seed attraction dominates the local probability.
 - **Beyond radius d_i :** The class 0 dominates.
 - **Smoothing effect:** Depending on α value, pixels can be affected more by its neighbors than by the radius rule.

The Theoretical Bottleneck: Intractable Likelihood

- **The Gibbs Measure:** The global probability of an image configuration ω is governed by:

$$P(\omega) = \frac{1}{Z(\theta)} \exp\{-U(\omega)\}$$

- **The Computational Bottleneck:** The term $Z(\theta)$ is the **normalizing constant** (partition function) summed over all lattice configurations Ω :

$$Z(\theta) = \sum_{\omega \in \Omega} \exp\{-U(\omega)\}$$

- For a small 50×50 grid and $K = 3$ colors, this requires 3^{2500} terms, rendering standard Maximum Likelihood impossible.

3. Hidden & Hierarchical Models

The Major Evolution: The Hidden Model (HMRF)

- **The Latent Field:** In segmentation practice, the label field Z (the true texture) is **latent/hidden**.
- **Observed Intensities:** We only observe the noisy, continuous intensities Y .
- **Observation Equation (Emission):** Conditionally on the latent label $Z_i = k$, the intensity Y_i is Normal:

$$Y_i \mid Z_i = k \sim N(\mu_k, \sigma_k^2)$$

- **Normal Mixture:** Marginalizing over Z_i , Y_i forms a spatial mixture of Gaussians:

$$f(y_i) = \sum_{k=0}^{K-1} P(Z_i = k) \phi(y_i; \mu_k, \sigma_k^2)$$

Hierarchical Bayesian Formulation

- **Full Joint Distribution:** Treating Z , parameters θ , and seed coordinates S as random variables:

$$P(Y, Z, \theta) = P(Y \mid Z, \mu, \sigma^2) P(Z \mid \alpha, \beta, \delta, S) \times \\ P(\alpha) P(\beta) \prod_{j=1}^M P(s_j) P(\delta_j) \prod_{k=0}^{K-1} P(\mu_k) P(\sigma_k^2)$$

- **Prior Distributions:**

- **Field (Gamma):** $\alpha \sim \text{Gamma}(a_\alpha, b_\alpha)$, $\beta \sim \text{Gamma}(a_\beta, b_\beta)$.
- **Means (Normal):** $\mu_k \sim N(m_k, \tau_k^2)$ (anchored via K-means).
- **Variances (Inverse-Gamma):** $\sigma_k^2 \sim \text{IG}(a_\sigma, b_\sigma)$.
- **Geography:** Seed position $s_j \sim \text{Unif}(L)$ discretized on the grid.

3. Estimation & Algorithms

The Metropolis-within-Gibbs Algorithm

- ➊ **Step C (Gibbs - Latent Field):** Sample $Z_i^{(t)}$ conditioned on its neighbors and the observed intensity Y_i .
 - ➋ **Step D (Gibbs - Noise):** Sample new (μ_k, σ_k^2) directly from closed-form analytical distributions.
 - ➌ **Step A (Metropolis-Hastings - Field):** Propose (α, β, δ) via a random walk, evaluated over the **Pseudolikelihood (PL)** or user-chosen priors.
 - ➍ **Step B (Metropolis-Hastings - Seeds):** Propose discrete 1-pixel steps for the coordinates (x, y) of the seeds.
- **The Local Gibbs Trick:** Global normalizing constant $Z(\theta)$ cancels out locally.

$$P(Z_i = k \mid Y_i, Z_{-i}, \theta) = \frac{\phi(y_i; \mu_k, \sigma_k^2) \exp\{\eta_i(k)\}}{\sum_{\ell} \phi(y_i; \mu_{\ell}, \sigma_{\ell}^2) \exp\{\eta_i(\ell)\}}$$

Full Conditionals and Conjugacy

- **Updated Posterior Mean (μ_k):** The full conditional under conjugate priors is a Normal:

$$\mu_{\text{post}} = \frac{\frac{m_k}{\tau_k^2} + \frac{\sum_{i:Z_i=k} y_i}{\sigma_k^2}}{\frac{1}{\tau_k^2} + \frac{n_k}{\sigma_k^2}}$$

- **Updated Posterior Variance (σ_k^2):** Follows an Inverse-Gamma:

$$a_{\text{post}} = a_{\sigma} + \frac{n_k}{2}$$

$$b_{\text{post}} = b_{\sigma} + \frac{1}{2} \sum_{i:Z_i=k} (y_i - \mu_k)^2$$

Code Engineering & Stability Challenges

- **Log-Sum-Exp Trick:** To prevent catastrophic underflow/overflow:

$$\log \sum_j \exp(x_j) = x_{\max} + \log \sum_j \exp(x_j - x_{\max})$$

- **Preventing Label Switching:** We enforce strict ordering of the means ($\mu_0 < \mu_1 < \dots$) at each Gibbs iteration.
- **Rcpp for Real-World Grids:** Recreated the local neighborhood and distance calculations in **C++ via Rcpp** to make iterations highly efficient.

4. Results & Application

Simulation Results

- **Parameter Recovery:** MWG algorithm shows rapid convergence after a short burn-in period.
- **Credible Intervals:** True spatial and geographic parameters are captured within 90% Bayesian credible intervals (5% and 95% quantiles).
- **Geographic Trajectory:** Seed positions walk stochastically and converge exactly on their true coordinates.

Image Segmentation Application

- **Texture Segmentation:** Without spatial priors, simple mixtures produce extremely noisy results.
- **Our Model:** The latent field with seeds creates **smooth regions**, preserving concentric attraction near seeds.
- **Visual Comparison:**
 - 1 Noisy Image $Y \rightarrow$ heavy noise.
 - 2 Our Model $\rightarrow$ clean, mathematically consistent shapes.

Conclusion and Next Steps

- **Key Conclusions:** The MWG pipeline makes inference in geographically complex HMRFs viable and robust.
- **Efficiency:** Estimation via Pseudoposterior balances computational cost and statistical precision.
- **Future Work:**
 - IDK